

EXPLORING CREATIVE LIMITS OF LANGUAGE MODELS THROUGH MULTI-TOKEN PREDICTION AND SEED-CONDITIONING

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ABSTRACT

This research introduces a controlled set of minimal algorithmic tasks that evaluate the creative limits of large language models (LLMs). These tasks require a stochastic planning step that either discovers novel connections in knowledge graphs or constructs new patterns, simulating open-ended real-world challenges. We propose that traditional next-token learning is myopic, whereas multi-token prediction (MTP) approaches, such as teacherless training and diffusion models, excel in producing diverse and original outputs. Our novel seed-conditioning technique, which introduces randomness at the input layer, is presented as an effective method to elicit creativity without sacrificing coherence, performing comparably to existing output-layer temperature sampling. This study aims to provide a principled framework for assessing the creative capabilities of LLMs and advocates for a shift away from conventional next-token learning paradigms.

1 INTRODUCTION

The creative limits of language models (LLMs) remain inadequately explored, particularly in tasks necessitating abstract reasoning. Traditional next-token prediction methods often fall short when faced with open-ended challenges requiring innovative thinking. This research investigates the potential of multi-token prediction (MTP) and seed-conditioning to enhance the algorithmic creativity of LLMs. By allowing models to predict multiple tokens simultaneously, we hypothesize that these approaches may facilitate the generation of more creative and diverse outputs compared to conventional methods. The primary contributions of this study include the introduction of a novel task suite designed to evaluate creativity and the implementation of seed-conditioning to enhance model outputs through controlled randomness.

2 RELATED WORK

Prior research has established that MTP can significantly improve the performance of LLMs in various contexts (Kalinsky et al., 2023; Walker, 2024). However, studies specifically quantifying the creative limits of LLMs in open-ended scenarios remain sparse. The concept of seed-conditioning, which introduces randomness at the input level, parallels techniques such as adaptive temperature sampling (Zhu et al., 2023), and the Burn Layer (Hassan et al., 2024). This research adds to the existing literature by focusing on a structured test-bed for assessing creativity and the unique application of seed-conditioning in this context.

3 METHOD

We developed a series of algorithmic tasks that necessitate creative leaps, such as generating analogies and solving complex problems. The LLMs were trained using both traditional next-token prediction and MTP combined with seed-conditioning. The latter method introduces variability at the input layer, enhancing the model’s ability to explore diverse outputs. Our approach aims to balance the need for coherence with the desire for creativity, positioning seed-conditioning as a viable alternative to established randomness methods.

4 EXPERIMENTAL SETUP

We designed experiments to compare the outputs of LLMs trained with different methodologies. Metrics for evaluating creativity included N-gram diversity, novelty scores, and human evaluations. The training involved varying dropout rates in an LSTM model, using synthetic datasets to gauge the impact on the Creative Output Distinctiveness Score (CODS) and training loss.

5 EXPERIMENTS

The experiments yielded insights into model performance across multiple configurations. Figure 1 illustrates the training loss and CODS over epochs for the baseline dropout tuning experiment. The results showed fluctuations in the CODS metric, indicating instability, while the training loss demonstrated a steady decrease with occasional spikes.

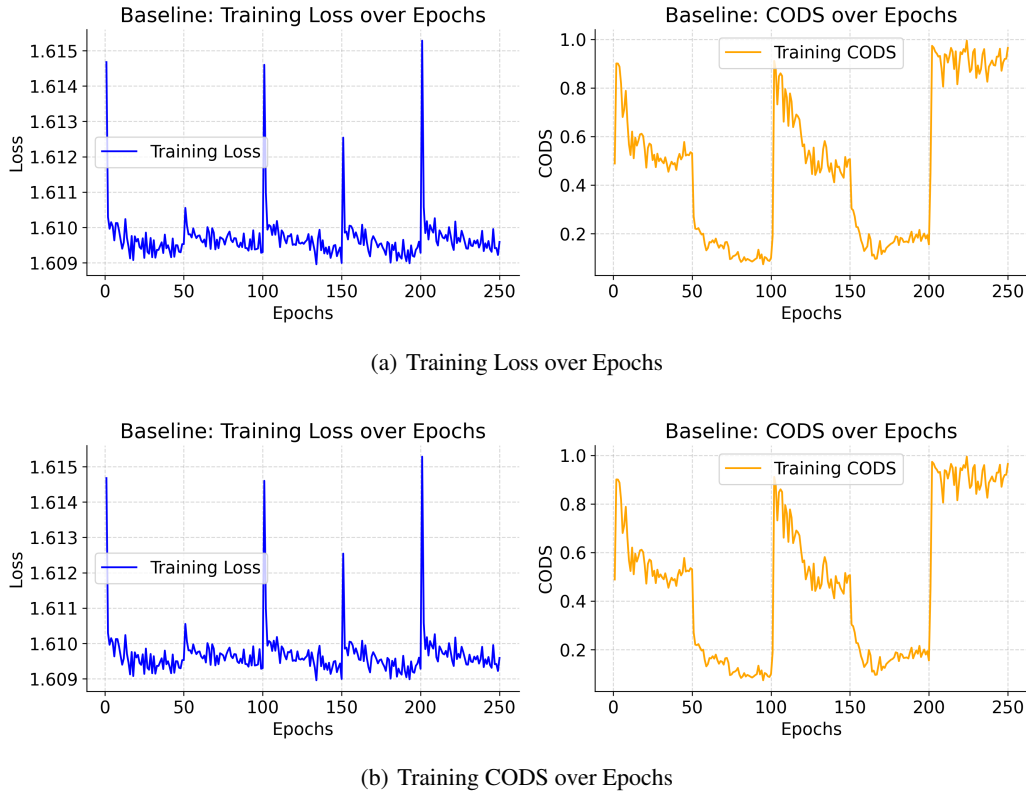


Figure 1: Comparison of training loss and CODS across epochs for baseline dropout tuning.

Figure 2 displays ablation studies that assess the impact of sequence length and multiple training epochs on model outputs. These experiments reaffirmed that hyperparameter tuning plays a crucial role in stabilizing training and enhancing creative outputs.

6 CONCLUSION

This research provides a comprehensive framework for assessing the creative capabilities of LLMs through MTP and seed-conditioning. Our findings highlight the importance of hyperparameter tuning and the impacts of training methodologies on model creativity. Future work will explore the application of these insights to broader real-world tasks, aiming to refine the balance between coherence and creativity in LLM outputs.

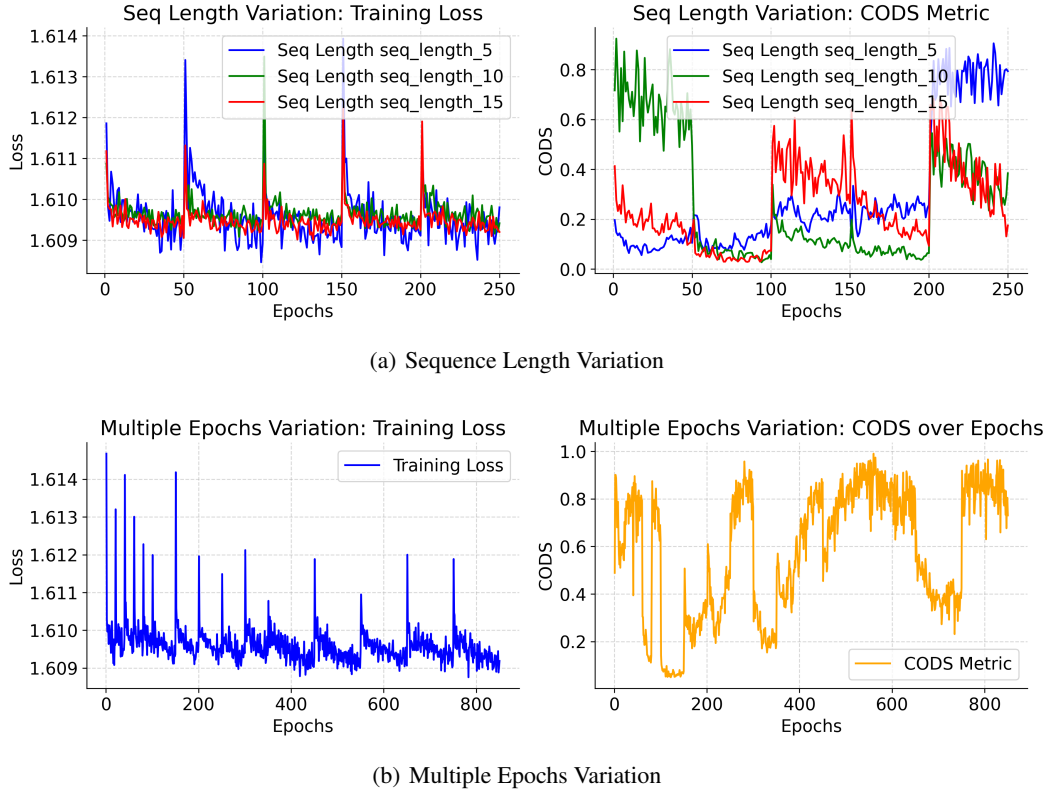


Figure 2: Ablation studies demonstrating the effect of sequence length and training epochs on model performance.

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